NORTH SOUTH UNIVERSITY

The First Private University in Bangladesh

Date: March 25, 2026

To: Graduate Admissions, University of Alberta

I understand that Mr. Jubair Ibn Omar Hamza has applied for admission to the Master of Engineering program in Electrical and Computer Engineering (Integrated Circuits and Systems) at the University of Alberta.

I have known Jubair for approximately two years at North South University, where I have supervised him closely as a Research Assistant. I have had the opportunity to observe his development as an exceptionally capable and disciplined young engineer with strong potential for advanced graduate studies. Based on these interactions, I consider Mr. Jubair to be a highly capable and well-prepared candidate for graduate study in Electrical and Computer Engineering.

Mr. Jubair completed his B.Sc. in Electrical and Electronic Engineering with Summa Cum Laude distinction, reflecting consistent excellence across core engineering courses. He is currently pursuing his M.Sc. in Electrical and Electronic Engineering, maintaining a strong academic record, and has further strengthened his analytical background through a Post Graduate Diploma in Data Science. This combination of formal engineering training and data-driven methodology enables him to approach complex technical problems with both physical insight and computational rigor.

In my research group, Mr. Jubair has contributed substantially to interdisciplinary work involving electromagnetic systems, signal processing, and computational modeling. His work on Pulsed Electromagnetic Field (PEMF) systems required him to design and implement signal processing pipelines using Python and MATLAB, conduct electromagnetic and multiphysics simulations using COMSOL, and analyze how modeling assumptions influence experimental interpretation. He has demonstrated a strong ability to understand governing equations, translate them into computational frameworks, and critically evaluate simulation results rather than relying on software tools alone.

A notable strength of Mr. Jubair is his ability to integrate theory, computation, and practical implementation. In addition to modeling and analysis, he has participated in the development of hardware–software systems using Raspberry Pi platforms, where he worked on sensor interfacing, data

acquisition, and real-time signal processing. Through this work, he developed a solid understanding of system-level considerations such as signal integrity, noise, sampling constraints, synchronization, and overall system reliability—skills that are directly relevant to integrated circuits and systems applications.

Importantly, his contributions were made within the framework of ongoing funded research activities, where he gained experience working with structured objectives, timelines, and technical deliverables. This exposure strengthened his research discipline, documentation practices, and ability to contribute effectively within organized engineering projects.

Beyond his technical contributions, Mr. Jubair exhibits strong professional discipline and problem-solving ability. He engages actively with relevant literature, asks well-formulated technical questions, and approaches complex tasks in a systematic and methodical manner. When faced with challenges, he demonstrates persistence and thoughtful troubleshooting. His experience as a Graduate Teaching Assistant and Laboratory Instructor further highlights his clarity in communication, reliability, and ability to work effectively in team-based academic environments.

Based on my experience mentoring a wide range of students in engineering and applied sciences, I rank Mr. Jubair among the top 5% I have supervised. He combines strong academic preparation, computational proficiency, and a practical engineering mindset. I am confident that he will succeed in the Master of Engineering program at the University of Alberta and contribute positively to its academically rigorous and application-oriented environment.

Sincerely,



Zaid Bin Mahbub, PhD
Associate Professor
Email: zaid.mahbub@northsouth.edu
Phone: +8801646984207
Department of Mathematics and Physics
North South University